Liquidity-Aware Smart Execution for Desk Efficiency

Microstructure Signals Across Manual Trading and Peak Algo

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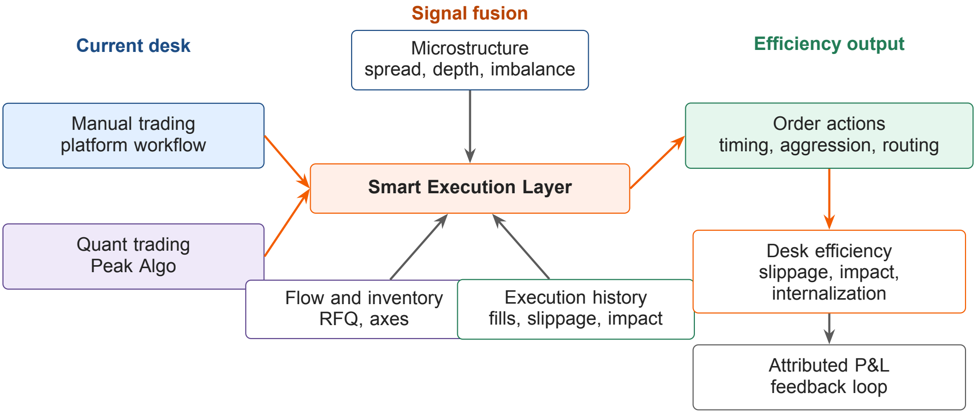
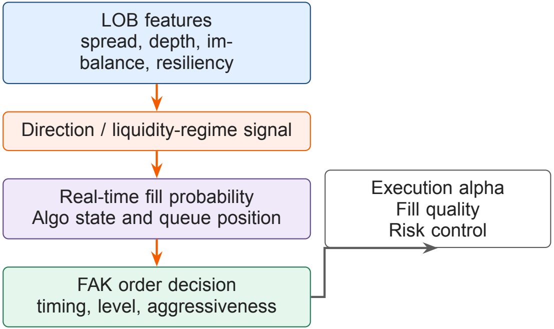
Smart Execution as a Desk Efficiency Layer



FAK Order Execution Alpha



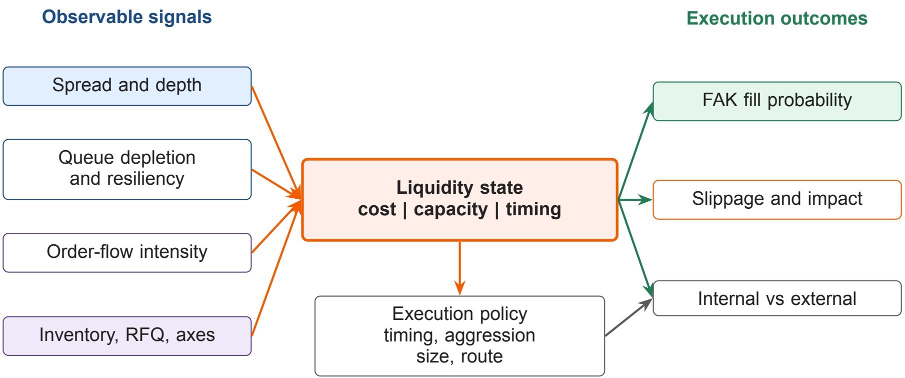
Case 1: favorable direction

- Direction supports the order side.
- Expected fill probability is higher.
- Execution alpha: lower slippage and lower market impact.

Case 2: adverse direction

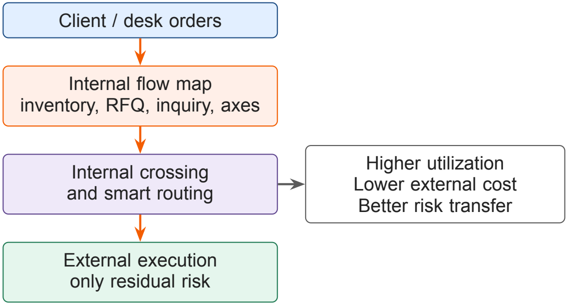
- Direction works against the FAK order.
- Revise fill expectation and aggressiveness.
- Improve success rate and reduce unhedged risk.

Liquidity is the True State Variable



Source: Lehalle and Laruelle; Cont and de Larrard; Bacry et al.; LOBFrame / LOB benchmark studies.

Microstructure for Internal Flow



Evidence-backed plan

- Segment flow by liquidity regime before routing.
- Prioritize internal crossing before paying spread and impact.
- Use queue and order-flow signals for residual external execution.
- Validate by internalization uplift and residual external cost.

How We Save 0.01 bp Per Trade

Base-case value

90M

annual P&L target from 0.01 bp saving

- 2025 flow: 45tn.
- Average duration: 2 years.
- $45tn \times 2 \times 0.01bp \approx 90M.$

Where the saving comes from

- **Internal crossing:** avoid spread and external market impact.
- **Timing:** trade when depth and resiliency are favorable.
- **Adverse-selection control:** avoid passive fills when direction is unfavorable.

How we prove it

- Slippage vs. arrival / decision price.
- Market impact after execution.
- Internalization uplift and residual external cost.